

PROJECT REPORT: Pizza Sales Performance & Inventory Optimization

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Date: 30/06/2026

Tools Used: Python (Pandas), SQL (MySQL), Power BI, Excel

1. Executive Summary

This project provides a 360-degree operational analysis for "Maven Pizza Challenge" to improve profitability and streamline kitchen operations. By analyzing over 20,000 orders, we identified key sales drivers, peak periods, and inventory requirements.

Key Highlights:

- **Total Annual Revenue:** ~\$817,860 (Approx based on dataset).
 - **Busiest Periods:** Lunch (12:00 PM) and Dinner (6:00 PM).
 - **Top Pizza:** Thai Chicken Pizza (Highest Revenue).
 - **Strategy Recommendation:** Phasing out XXL sizes and increasing staffing for the Friday evening rush.
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2. Business Problem Statements

The store manager required answers to four critical areas:

1. **Sales Trends:** When is the store most overwhelmed?
 2. **Product Performance:** Which pizzas drive the most profit vs. volume?
 3. **Inventory Management:** What are the daily stocking requirements for raw ingredients?
 4. **Pricing Strategy:** Does the current size-to-price ratio maximize revenue?
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3. Data Architecture & Tech Stack

The project utilized a relational database structure consisting of four tables: orders, order_details, pizzas, and pizza_types.

- **Python:** Used for the "Ingredient Explosion." We broke down the comma-separated ingredient list into a flat file to calculate exact raw material demand.

- **SQL:** Used to perform complex aggregations, identify peak hours, and calculate Average Order Value (AOV).
- **Power BI:** Used for the final delivery. Created an interactive dashboard with slicers for Day, Month, and Category.

4. Key Findings & Insights

A. Temporal Trends (The "When")

- **Peak Hours:** Sales spike sharply at **12:00 PM** and **6:00 PM**.
- **Peak Days:** Friday and Saturday are the highest-volume days.
- **Seasonality:** July recorded the highest number of orders, while February was the slowest month.

B. Product Performance (The "What")

- **Revenue Leaders:** Thai Chicken and BBQ Chicken pizzas contribute the most to the bottom line.
- **Quantity Leaders:** Classic Deluxe and Pepperoni are the most frequent orders.
- **Underperformers:** The **Brie Carre Pizza** ranks last in both revenue and quantity, suggesting it should be removed or rebranded.

C. Pricing & Size Analysis

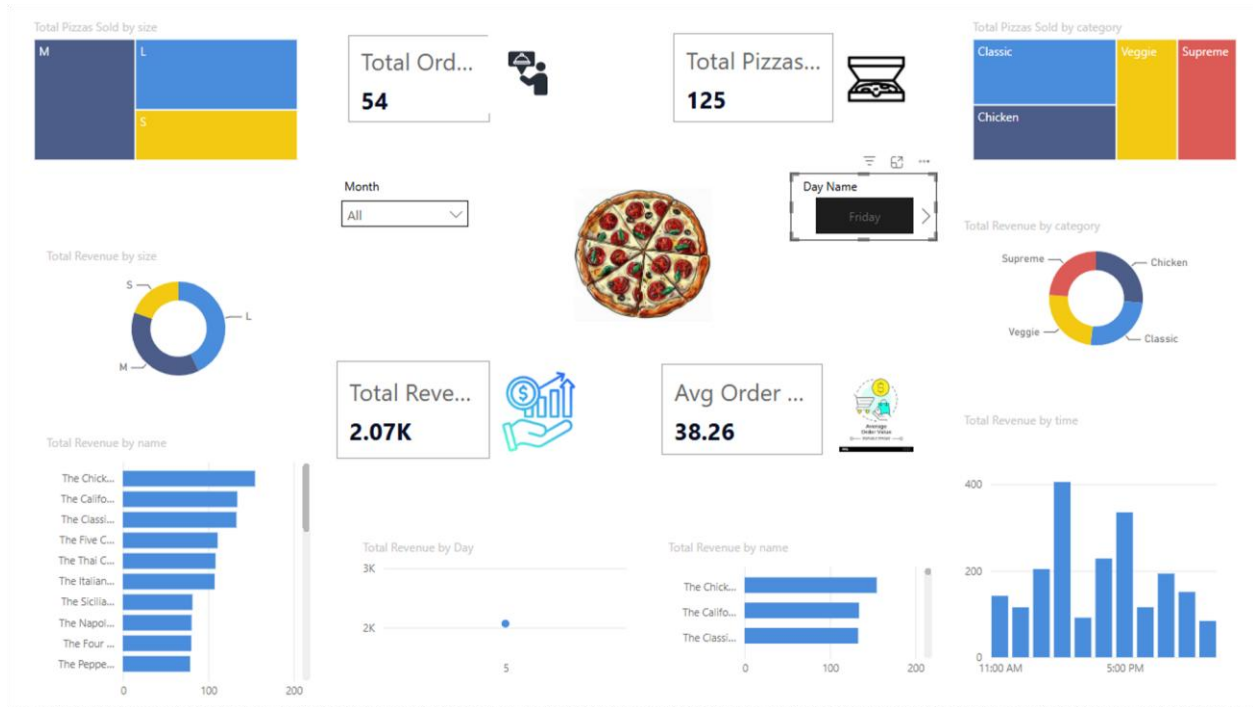
- **The "Large" Driver:** Large (L) pizzas account for nearly **45%** of total revenue.
- **Size Rationalization:** XL and XXL pizzas contribute less than **2%** of total sales, yet require unique packaging and storage space.

5. Strategic Recommendations

1. **Staffing Optimization:** Increase kitchen staff by 20% during the 5:00 PM – 8:00 PM window on Fridays and Saturdays.
2. **Inventory Strategy:** Establish a "Daily Standing Order" for Garlic, Tomatoes, and Red Onions, as these appear in over 80% of all pizzas sold.
3. **Menu Engineering:** Discontinue the XXL size. Promote a "Medium to Large" upsell deal to increase the Average Order Value (AOV).

4. **Marketing:** Launch "Lunch Specials" specifically between 1:00 PM and 3:00 PM on weekdays to bridge the gap between the lunch and dinner peaks.

6. Appendix: Dashboard & Code



SQL Sample (Top Pizzas by Revenue):

```
SELECT name, SUM(quantity * price) AS total_revenue
FROM Cleaned_pizza_sales
GROUP BY name
ORDER BY total_revenue DESC LIMIT 5;
```

Python Sample (Ingredient Extraction):

```
# Exploding ingredients for inventory analysis
```

```
Ingredients
```

```
df = master_df.assign(ingredients=master_df['ingredients'].str.split(', ')).explode('ingredients')
```